THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $(x + 10)^\circ$ and $(2x - 4)^\circ$ are supplementary, x equals: 1

(a) 68 **(b)** 63 **(c)** 58 **(d)** 53

2. If $l \parallel m$ and two co-interior angles are 68° and $(x)^\circ$, then $x =$ 1

(a) 112° **(b)** 132° **(c)** 102° **(d)** 122°

3. The angle which is 2 times its complement is: 1

(a) 50° **(b)** 80° **(c)** 60° **(d)** 70°

4. Half of the supplement of 40° is: 1

(a) 90° **(b)** 60° **(c)** 70° **(d)** 80°

5. The angle bisector of a 70° angle makes each part: 1

(a) 45° **(b)** 35° **(c)** 25° **(d)** 55°

6. A transversal makes alternate interior angles of 65° and 65°. The two lines are: 1

(a) parallel **(b)** intersecting **(c)** cannot say **(d)** perpendicular

7. A transversal makes co-interior angles in the ratio $1 : 2$ on parallel lines. The smaller angle is: 1

(a) 60° **(b)** 70° **(c)** 50° **(d)** 80°

8. If $(x + 20)^\circ$ and $(2x - 8)^\circ$ are supplementary, x equals: 1

(a) 61 **(b)** 51 **(c)** 56 **(d)** 66

9. If $l \parallel m$ and two co-interior angles are 87° and $(x)^\circ$, then $x =$ 1

(a) 103° **(b)** 113° **(c)** 93° **(d)** 83°

10. The angle which is 5 times its complement is: 1

(a) 75° **(b)** 65° **(c)** 95° **(d)** 85°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the angle that is 20° less than its supplement. 2

12. Find x if $(3x - 10)^\circ$ and $(x + 20)^\circ$ are vertically opposite angles. 2

13. Is $l \parallel m$ if a transversal gives corresponding angles 75° and 75°? Give reason. 2

14. Find the angle that is 40° less than its supplement. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A transversal cuts parallel lines. Interior angles on the same side are $(2x + 10)^\circ$ and $(3x + 15)^\circ$. Find x and the two angles. **3**

16. In a pair of intersecting lines, the angles are $(x + 15)^\circ$, $(2x - 9)^\circ$ forming a linear pair. Find x and the vertically opposite angles. **3**

17. An angle is 6° more than twice its complement. Find the angle and its complement. **3**

OR

A transversal cuts parallel lines. Interior angles on the same side are $(2x + 5)^\circ$ and $(3x + 20)^\circ$. Find x and the two angles. **3**

18. In a pair of intersecting lines, the angles are $(x + 30)^\circ$, $(2x - 6)^\circ$ forming a linear pair. Find x and the vertically opposite angles. **3**

OR

An angle is 15° more than twice its complement. Find the angle and its complement. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Lines $l \parallel m$, transversal t. $\angle 1 = (2x + 10)^\circ$, $\angle 2 = (3x - 20)^\circ$ are alternate interior angles; $\angle 3$ is the co-interior angle of $\angle 1$. Find x, $\angle 1$, $\angle 2$ and $\angle 3$. **5**

OR

Around a point, angles are x°, $2x^\circ$, $3x^\circ$ and 30°. Find x and each angle, and say which are acute, right or obtuse. **5**

20. Lines $l \parallel m$, transversal t. $\angle 1 = (2x + 15)^\circ$, $\angle 2 = (3x - 10)^\circ$ are alternate interior angles; $\angle 3$ is the co-interior angle of $\angle 1$. Find x, $\angle 1$, $\angle 2$ and $\angle 3$. **5**

OR

Around a point, angles are x°, $2x^\circ$, $3x^\circ$ and 60°. Find x and each angle, and say which are acute, right or obtuse. **5**

END OF QUESTION PAPER — Total: 40 Marks

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